Does Syntactic Structure Make Language Models More Robust to Noise?

LSTM vs. RNNG Subject–Verb Agreement Under Training-Data Corruption

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DSC 291: Language Models as Cognitive Models
UC San Diego

June 2026

Outline

1. Motivation
2. Background
3. Methods
4. Results
5. Discussion

Motivation: robust grammar from noisy input

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Research questions

1. TODO: can injected SVA errors delay/prevent the hierarchical rule?
2. TODO: is the linear rule equally learnable under noise?
3. TODO: do RNNG vs. LSTM differ in robustness?

Subject–verb agreement: hierarchical vs. linear

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Attractors as the diagnostic case

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Prior work

- TODO

Noise injection: five corrupted corpora

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Leak-free evaluation protocol

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Models and metrics

- TODO

Accuracy degrades with noise

TODO: `acc_vs_noise.png`

The attractor gap widens with noise

TODO: `attractor_gap.png`

Accuracy by attractor presence

TODO: `acc_by_attractor.png`

Learning trajectories: delay vs. prevention

TODO: `acc_trajectory.png`

Does structure buy robustness?

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Takeaways and future work

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Thank you!

Questions?